CCC-C-460C
January 21, 1976
SUPERSEDING
Fed. Spec. CCC-C-460B
June 23, 1971

FEDERAL SPECIFICATION

CLOTH, FLANNEL, COTTON (HEAVY, FOR TABLE FELTS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of cotton flannel cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-1134 - Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Test Method Std. No. 191 - Textile Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issues, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charges from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 8305

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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article approval. The requirement for first article will be as specified by the procuring activity (see 4.2 and 6.2).

3.3 Material.

3.3.1 Yarn. The yarn shall be made of cotton which has been carded, drawn, and spun into singles for both the warp and filling (see 4.4).

3.4 Physical requirements. The finished cloth shall conform to the requirements in Table I when tested as specified in 4.4.

TABLE I. Physical requirements

Weight oz. per sq. yd. (min.)	Yarns per in. (min.)		Breaking strength (min.) pounds	
	Warp	Filling	Warp	Filling
12.5	66	42	60	30

3.4.1 Width. The width of the cloth shall be as specified and shall be the minimum acceptable width inclusive of selvages (see 6.2).

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3.4.2 Weave. The cloth shall have a filling reversible (filling backed) weave, i.e., one warp and two filling systems. One filling shall weave on the back and one filling on the face of the cloth. The basic weave shall be 3 down, 1 up regular or broken twill and shall be the same when viewed on either the face or back side.

3.5 Finish. The cloth shall be unbleached (natural) and shall have a full nap on both sides when inspected as specified in 4.4.

3.6 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be put-up on rolls as specified in PPP-P-1134. Rolls shall be in one continuous piece of not less than 25 yards for 48 and 54 inch widths and not less than 30 yards for 64 and 72 inch widths.

3.7 Fiber identification. Each piece of cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.8 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. The first article submitted in accordance with 3.2 shall be visually inspected for finish, construction, and appearance and shall be tested for physical properties in accordance with the test methods specified in 4.4.

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

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4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item. Examination of the end item shall be in accordance with 4.3.2.1 through 4.3.3.

4.3.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on one side only. The side examined, however, shall be reversed so that alternate rolls are examined on the opposite side. When the roll contains less than 100 yards, the entire roll shall be examined. When the roll contains 100 yards or more, only 100 yards shall be examined. All defects as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distances (3 feet), shall be scored and assigned demerit points as listed in 4.3.2.2. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 32.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more rolls exceeds 48.0 points. If one individual roll exceeds 48.0 points per 100 square yards, a second sample equal in size to the first shall be selected for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds 48.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.3.2.2 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects when present, shall be scored four points for each yard in which they occur:

Overall uncleanness
Objectionable odor
Width less than specified
Color other than unbleached (natural)

4.3.2.3 Examination for length.

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4.3.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.3 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for compliance with the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3.4 Examination for preparation of delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1134, to determine whether packaging, packing and marking comply with Section 5 requirements of this specification.

4.4 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II shall be followed. The physical and chemical values specified in Section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be one yard full width of the finished cloth. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of one yard each. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The sample size shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 up and including 22,000	3
22,001 and over	5

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TABLE II. Test methods

Characteristic	Requirement reference	Test method
Fiber identification	3.3.1	1200 1/
Yarn Ply	3.3.1	Visual 2/
Weave	3.4.2	Visual 2/
Weight	3.4	5041
Yarns per inch	3.4	5050
Breaking strength	3.4	5100
Finish	3.5	1/

1/ Unless otherwise specified, a supplier's certificate of compliance shall be submitted and will be accepted for this characteristic.
2/ One determination shall be made on each sample unit and the result reported as "pass" or "fail."

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use as table felts.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Width of cloth required (see 3.4.1).
(c) Length of rolls if other than specified (see 3.6).
(d) Selection of applicable levels of put-up, packaging, and packing (see 5.1 and 5.2).
(e) When first article is required (see 3.2)

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6.3 Standard sample. For access to standard sample, address the procuring officer issuing the invitation for bids (see 3.1)

MILITARY CUSTODIANS:
Navy - SA
Air Force - 82

Review activity:
Air Force - 82

User activities:
Navy - MC, CG, MS

Preparing activity:
Navy - SA

CIVIL AGENCIES INTEREST:
GSA-FSS
VA- DMS

Project No. 8305-0348

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DOCUMENT IDENTIFIER AND TITLE
CCC-C-460C CLOTH, FLANNEL, COTTON (HEAVY, FOR TABLE FELTS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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